SOLUTIONS

SE2228 ANAYSIS AND DESIGN OF ALGORITHMS
QUIZ #1 EXAM - SECTION II

ID: **Name:** **(60 min.)**
22.10.2020

1. What is the complexity of the following code segment? Explain your answer.

```
//All variables are already declared
for ( i= 0; i < n;i++) {
        temp=n; while(temp>0)
        temp=temp/2;
}
```

$O(n \log n)$

Outer loop is repeated n times and the inner while loop is executed logn times

2. What does the following function compute?

```
int dce (int x){
        if (x!=0) { return x + dce(x-1); }
        else { return x; }
```

Sum of the numbers 1 to x

3. If t1(n) is O(f1(n)) , t2(n) is O(f2(n)) and t3(n) is O(f3(n)) then

What is the complexity of t1(n)+t2(n)+t3(n) ?

Max of the three

4. Order (in big O): $3^n + 5$, $n!$, $log(3^n)$, $n^3 + 43n^2 + 54n$, $3^{log(n)}$, 2^n

$\log(3^n)$, $3^{\log(n)}$, n^3+43n^2+54n, 2^n, 3^n+5, $n!$

5. Determine the value of the following sums:

a) $\sum_{i=5}^{i=20} 5i = 5\sum_{i=1}^{i=20} i - 5\sum_{i=1}^{i=4} i = 5\left(\frac{20.21}{2}\right) - 5\left(\frac{4.5}{2}\right) = 1000$

b) $\sum_{i=1}^{i=55} 4i + 5 = 4\sum_{i=1}^{i=55} i + \sum_{i=1}^{i=55} 5 = 4\left(\frac{55.56}{2}\right) + 5(55) = 6435$

6.

a) Find the complexity of T(n) = $\log 14n^n$

$O(n) = n \log n$

b) Show that T(n) = $5n^2 + 2n + 17$ is $\mathcal{O}(n^2)$

$5n^2+2n+17 <= c\, n^2$ Take $c=6$ then

$(c-5)n^2 >= 2n+17$ $n=2$ (minimum)

7. What is the solution of the following recurrence relation:

A(0) = 0 and A(n) = A(n-1) + 2n - 1

$[A(n-2) + 2(n-1) - 1] + 2n - 1$

$= A(n-2) + 2(n-1) + 2n - 2$

$= [A(n-3) + 2(n-2) - 1] + 2(n-1) + 2n - 2$

$= A(n-3) + 2(n-2) + 2(n-1) + 2n - 3$

Let generalize the relation

$= A(n-i) + 2(n-i+1) + \ldots + 2n - i$

let $i = n$

$= A(n-n) + 2(n-n+1) + \ldots + 2n - n$

$= 0 + 2 + 4 + \ldots + 2n - n$

$= 2(1+2+3\ldots n) - n$

$= 2\,\frac{n(n+1)}{2} - n = n^2$

8. Is recursion a good option if we have very large numbers of inputs and why?

No. Because a stack is created for storing local parameters and return addresses of recursive calls. Stack and its operations create overhead (Task time)